
Software Requirements Specification

for

DataBridge

Version 1.1.X approved

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Revision History

Name	Date	Reason For Changes	Version
Alec Howard	9/23/2026	Document Creation	1.0.X
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1. Introduction

1.1 Purpose

This document specifies the requirements for DataBridge, a browser-based bedside documentation system designed for rural hospitals operating on legacy MEDITECH platforms. It covers the full initial system, including bedside vitals entry, HL7v2 message generation, the simulated EHR receiver, and the security features that support them. This SRS does not cover a full EHR replacement, only the point of care documentation and interoperability workflow within the project's defined scope.

1.2 Document Conventions

Functional requirements are labeled **FR-#** and non-functional requirements **NFR-#**, numbered sequentially for traceability. Each requirement carries its own priority level (High/Medium/Low) rather than inheriting one from a broader category, since several individual requirements, particularly those related to security, need to be prioritized independently even when the larger feature they belong to is not urgent. Key technical terms, such as HL7v2, PHI, and RBAC, are defined at first use and kept consistent with the team's project glossary throughout.

1.3 Intended Audience and Reading Suggestions

This is intended for the development team, the project mentor and course evaluators. Developers should focus on the functional and Non- Functional requirements sections, since these define what the system must do and how it must behave. The mentor and evaluators assessing project scope and feasibility should begin with Product Scope and then review the Security Requirements sections specifically, as security was identified as a priority area during proposal feedback.

1.4 Product Scope

Data Bridge is an app based system that lets nurses enter vitals and short bedside notes on a tablet instead of on paper, converting each entry into an HL7v2 message sent to a simulated EHR receiver. The goal is to reduce the delay and error caused by the paper-based double documentation, a problem that particularly affects rural hospitals still running legacy EHR platforms. Scope is limited to point of care vitals, notes and the messaging workflow, with security (**RBAC, audit logging, encryption**) built in from the start.

1.5 References

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2. Overall Description

2.1 Product Perspective

Point of Care is a new, self-contained system rather than a follow-on to an existing product. It is not an existing product. It is not replacing a hospital's EHR; it functions as a lightweight front-end layer that sits between the nurse at the bedside and the hospital's existing EHR system. The product consists of three interconnected components: a tablet-facing frontend for data entry, a backend that validates and converts that data into an HL7v2 message, and a receiving interface that accepts the message. In a production deployment, that receiving interface would be the hospital's actual MEDITECH system; for this capstone, it is a simulated receiver built by the team, since the production access to a live hospital EHR is not available to a student project. The two systems interface through a single, well defined channel a standardized HL7v2 message meaning the front end and back end can be developed and evaluated independently of whichever system ultimately receives the data.

2.2 Product Functions

At a high level, the system allows a user to:

- Authenticate and access a role- appropriate view (nurse, charge nurse, or administrator)
- Select a patient from an active census
- Entered a structured bedside vitals (heart rate, blood pressure, temperature, respiratory rate, pain level) and a short observation note on a single-page interface
- Submit the entry for backend validation
- Have the entry automatically converted into a standardized HL7v2 message and transmitted to the EHR receiver
- View incoming messages on the receiver side, confirming the entry was received correctly
- Have every action (view,edit, access attempt) recorded in an audit log

These functions fall into three related groups: bedside data capture (nurse-facing) interoperability processing (backend/message generation), and administrative oversight (dashboards, audit logs, access control) - detailed requirements for each are provided in Section 3

2.3 User Classes and Characteristics

- Nurse (primary user): Uses the system most frequently, during active shifts, document patient vitals and notes at the bedside. Assumed to have limited time per interaction and general but not necessarily deep technical comfort. The interface must be fast and simple to use under time pressure. Access is limited to their assigned patients.
- Charge Nurse (secondary user): Uses the system less frequently than a bedside nurse but requires a broader view. The full unit census rather than individual assigned patients to monitor overall patient status.
- Administrator: Uses the system least frequently, primarily for user management and reviewing security/audit information rather than clinical documentation. Assumed to have the highest level of system access and lowest need for bedside speed.

The nurse user class is the most important to satisfy, since the system's core value proposition faster, more reliable bedside documentation depends entirely on the nurse facing workflow being fast and easy to use.

2.4 Operating Environment

The system is a browser based web application, intended to run on a tablet (e.g iPad) at the bedside, though it also functions on a standard desktop browser for administrative and development use. It does not require native mobile app or app-store installation. The backend runs as a standard web service (Python with FastAPI) and does not require integration with a live hospital network for this capstone's scope; it communicates only with the team's own simulated EHR receiver. The system does not currently target a specific hospital's live network environment, since production access is outside this project's scope, as established in Section 1.

2.5 Design and Implementation Constraints

- **No production hospital access:** The team cannot access a real live MEDITECH system. As a result, all interoperability testing occurs against a simulated receiver built by the team, not the production system it represents.

- **Synthetic data only:** No real Protected Health Information (PHI) is used at any point in development or testing, consistent with the project's HIPAA- aligned but not HIPAA-required design.
- **Standard selection:** The system generates HL7v2 messages specifically, rather than the newer FHIR standard, because HL7v2 is the format legacy MEDITECH platforms actually use, this is a deliberate design constraint tied directly to the project's rural, legacy-system focus rather than a technical limitation.
- **Technology stack:** Backend implemented in Python using the FastAPI framework; frontend implemented using React or standard HTML/CSS/JavaScript. These choices were selected for team familiarity and development speed within the capstone timeline.
- **Security requirements:** Role-based access control, encrypted data in transit and at rest, and full audit logging are required design elements from the outset, per instructor feedback received during proposal review.
- **Team and timeline constraints:** The system is developed by a four-person student team within a single academic year, which limits the scope to point-of-care vitals and notes rather than full EHR feature set.

2.6 User Documentation

The team plans to deliver a short user guide covering the nurse-facing bedside workflow(login, patient selection, data entry, submission) and a brief administrator guide covering user management and audit log review. Documentation will be delivered as a written document (e.g., PDF or Word) accompanying the final system rather than as in-application help or an interactive tutorial, given the project's scope and timeline.

2.7 Assumptions and Dependencies

- It is assumed that a hospital contact (obtained through a team member's prior IT internship) will remain available to provide workflow input and where possible, example HL7v2 message formats; if this contact becomes unavailable, the system will still be built and validated against the public HL7v2 standard and synthetic data alone.
- It is assumed that no real hospital data will grant the team production system access or real patient data at any point during this capstone; the entire design is built around this assumption rather than treating it as a risk to be resolved,
- The project depends on third party open source libraries for HL7v2 message parsing and generation (e.g Python-H17), and assumes these libraries remain available and adequately documented for the team's implementation needs.
- The project depends on synthetic patient data generation tools (e.g., Synthea) remaining available and suitable for generating realistic bedside vitals data for testing,
- It is assumed that free-tier cloud hosting and development tools (as outlined in the project's Budget and Resources section) will remain sufficient for development and demonstration needs, given the project's minimal budget,

3. External Interface Requirements

3.1 User Interfaces

The DataBridge system shall provide a graphical user interface for point-of-care patient documentation. The Phase 1 prototype provides interfaces for nurse authentication, patient selection, and patient documentation.

3.1.1 Nurse Login Interface

The system shall provide a nurse login interface containing:

- DataBridge system title.
- Point-of-Care Patient Documentation title.
- Username input field.
- Password input field.
- **LOG IN** button.
- Authentication feedback.
- Demo-mode indication when synthetic data is being used.

The username field shall allow the nurse to enter their username. The password field shall conceal the entered password characters. The nurse shall select **LOG IN** to submit the credentials for authentication.

The Phase 1 prototype operates in demo mode using synthetic data.

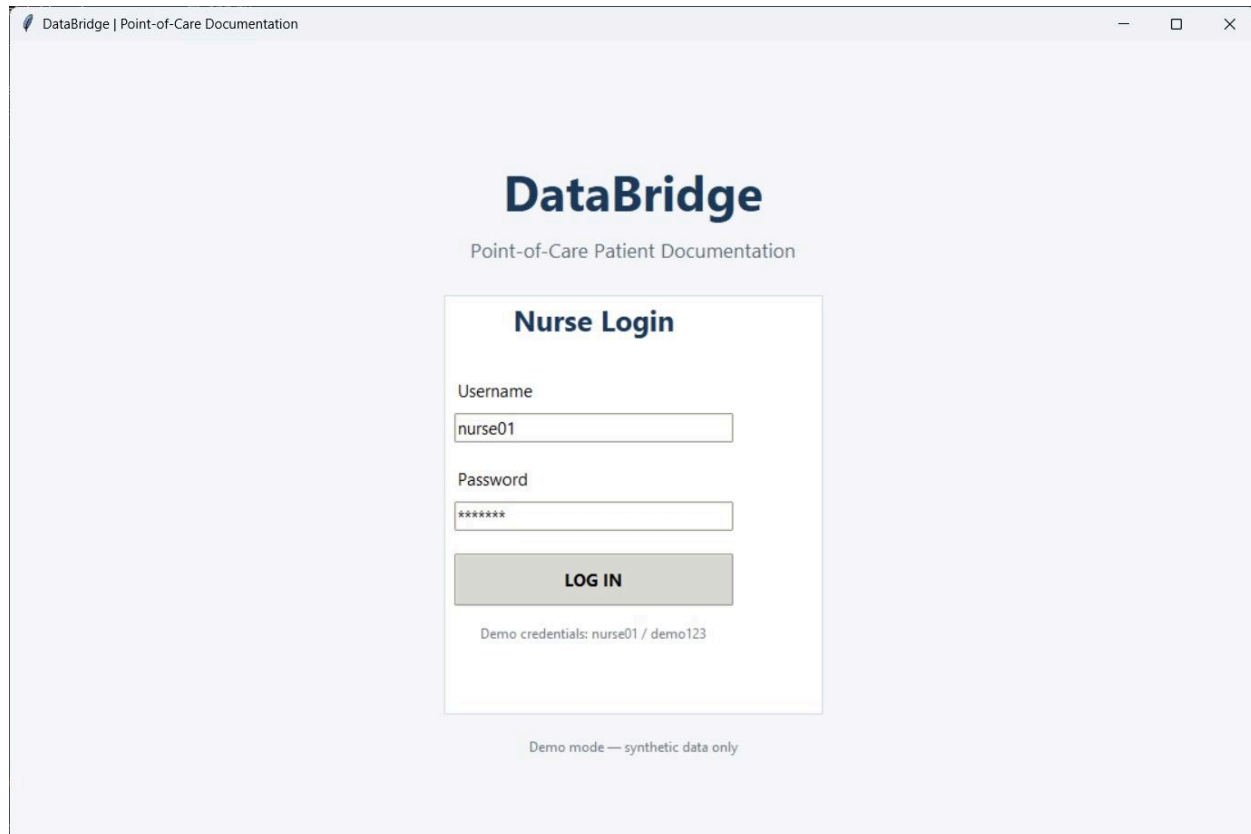


Figure 3.1.1: DataBridge Nurse Login Interface

3.1.2 Active Patient Census Interface

Following successful authentication, the system shall display an **Active Patient Census** interface. The patient census shall display patient records in a tabular format containing:

- Patient name and identifier.
- Room number.
- Patient status.
- Date of birth.

The interface shall allow the nurse to select a patient record to begin bedside documentation. The interface shall also display the currently authenticated user and their assigned role.

The Phase 1 prototype uses synthetic patient records for demonstration and testing.

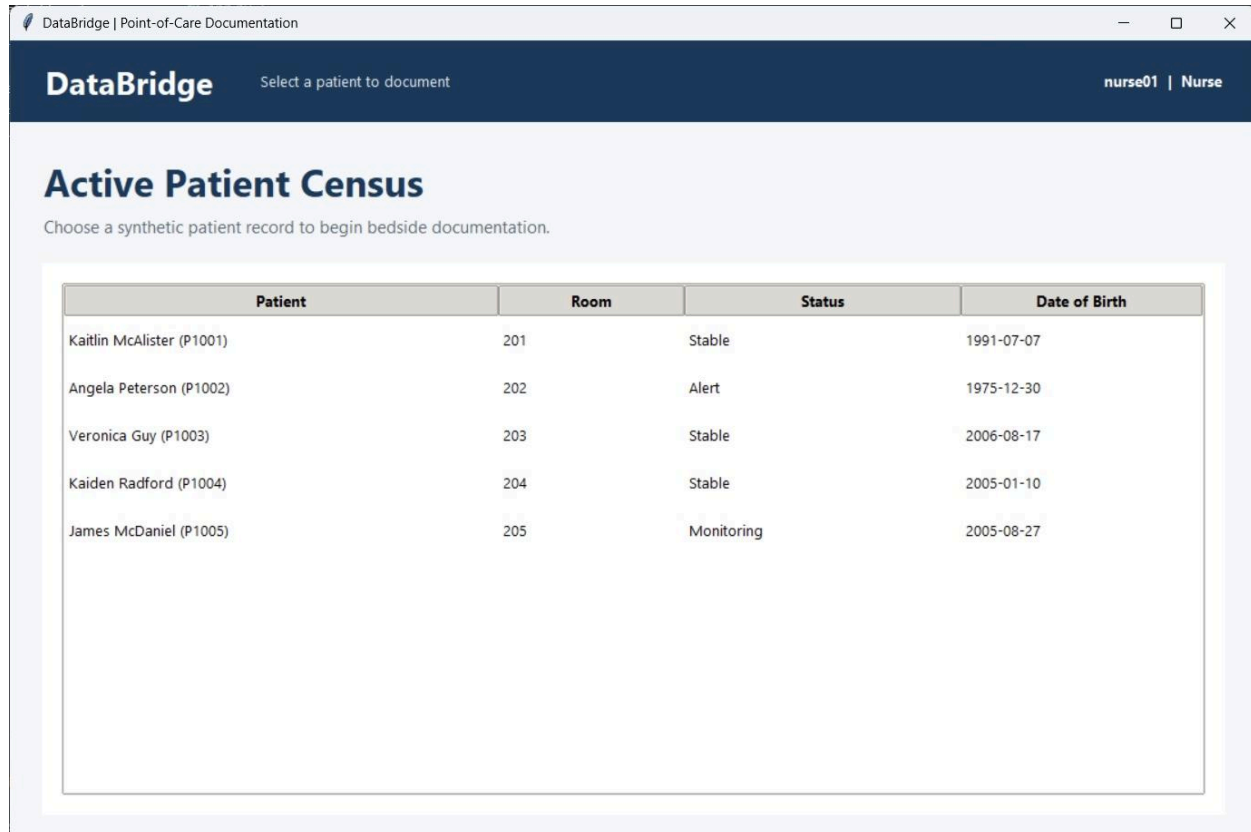


Figure 3.1.2: DataBridge Active Patient Census Interface

3.1.3 Patient Documentation Interface

After selecting a patient, the system shall display the patient's documentation interface.

The interface shall display:

- Patient name.
- Patient identifier.
- Room number.
- Patient status.
- Authenticated user and role.

The interface shall provide input fields for the following vital signs:

- Temperature in degrees Fahrenheit (°F).
- Heart rate in beats per minute (opm).
- Blood pressure in mmHg.
- Respiratory rate per minute.
- Oxygen saturation (SpO₂) as a percentage.

The interface shall provide text-entry areas for:

- Observations.
- Nursing Notes.

The interface shall provide a **BACK TO CENSUS** control that returns the user to the Active Patient Census.

The interface shall provide a **SUBMIT DOCUMENTATION** control for submitting the entered documentation.

The screenshot shows a web application window titled "DataBridge | Point-of-Care Documentation". The header bar is dark blue with the "DataBridge" logo on the left, "Patient P1001 • Room 201" in the center, and "nurse01 | Nurse" on the right. The main content area has a light blue background. At the top, the patient's name "Kaitlin McAlister" is displayed in large bold text, followed by "Room 201" and "Status: Stable". Below this, there are two main sections: "Vital Signs" on the left and "Observations" and "Nursing Notes" on the right. The "Vital Signs" section contains five input fields: Temperature (°F), Heart Rate (bpm), Blood Pressure (mmHg), Respiratory Rate (/min), and SpO₂ (%). The "Observations" section has a large text area, and the "Nursing Notes" section has a large text area. At the bottom of the interface, there are two buttons: "← BACK TO CENSUS" on the left and "SUBMIT DOCUMENTATION" on the right.

Figure 3.1.3: DataBridge Patient Documentation Interface

3.1.4 User Interface Navigation

The Phase 1 interface shall support the following workflow:

1. The nurse enters their username and password.
2. The nurse selects **LOG IN**.
3. The system displays the Active Patient Census.
4. The nurse selects a patient.
5. The system displays the patient's documentation interface.
6. The nurse enters vital signs, observations, and nursing notes.
7. The nurse selects **SUBMIT DOCUMENTATION**.

8. The system processes the submitted documentation and provides the appropriate submission status.

The user shall be able to return from the documentation interface to the patient census using the **BACK TO CENSUS** control.

3.2 Hardware Interfaces

An Apple Ipad will be the primary hardware device that will run the DataBridge application, and nurses or any other assigned employee will use the tablet to document at the point of care. Nurses or designated others will use the tablet to access the browser-based DataBridge application through which users can log into the application and select patients, enter vitals, make short observations, and document. The tablet will require an updated web browser and reliable internet connection to allow communication with the DataBridge backend. With a tablet, nurses will document patients' information at the point of care using an electronic method, as opposed to using paper-based documentation or having to use stationary computers.

3.3 Software Interfaces

The operation of the system will involve the authorized user inputting data about patients through the web interface of the tablet. The entered data will then be transferred through the API to the DataBridge backend. There, the data will undergo validation before processing. After the data has been processed and is ready for transmission, the DataBridge backend will transmit it to the HL7v2 message generator that will then generate a message and transfer it to the simulated EHR receiver.

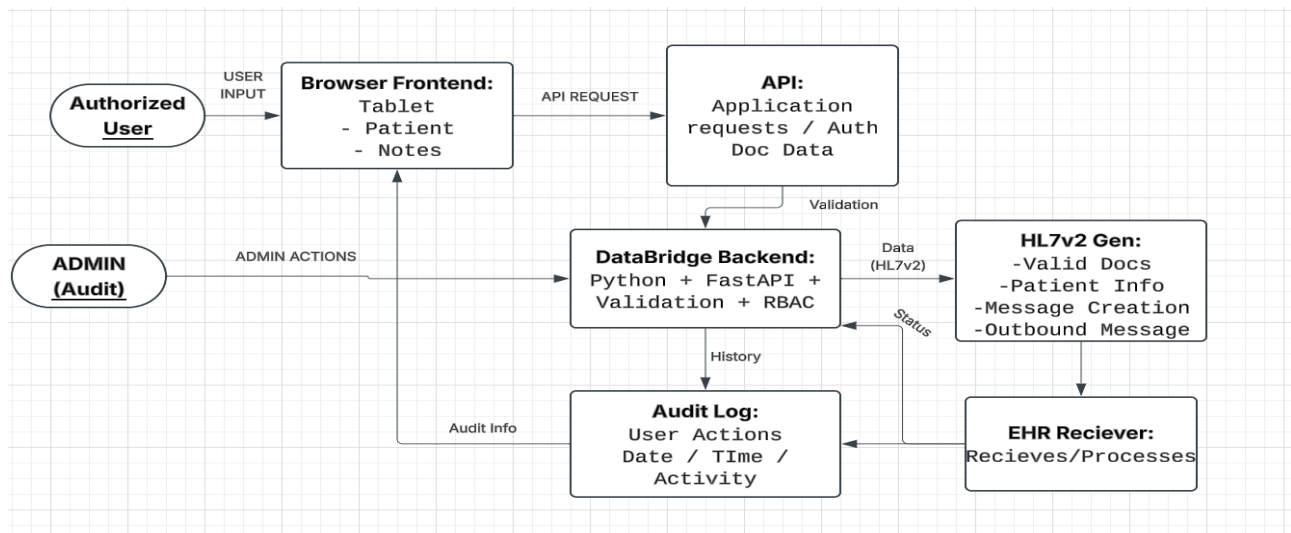


Figure 3.3.1: Software Interface Design

3.4 Communications Interfaces

DataBridge will be a web application that can be used in the browser, mostly in a tablet but also on desktop computers when necessary. Communication will occur between DataBridge Application and its backend using API requests over HTTP/HTTPS with information such as login credentials, patient choice, vital signs, bedside notes, and other documentation. Once the information is validated and processed, an HL7v2 message will be generated for the simulated EHR receiver. The use of HTTPS and encryption will allow it to securely transmit information between system components. Web communication will be used by the application to exchange information between the frontend and backend, whereas HL7v2 will be used to communicate with the simulated EHR.

4. System Features

This section follows the key features and capabilities of the DataBridge system. These features outline the way users will interface with the system and the way the system will perform patient data manipulation, verification, protection, and transmission. Key features of the primary workflow are: user authentication, authorization based on roles, bedside documentation, backend verification, HL7v2 message generation, and communication with the simulation of the EHR receiver. Other features are: audit logging, metadata extraction using OCR, and secure document transfer. Every feature is described by its purpose and importance, user-system interaction flow, and functional requirements defining what the system must be able to do. The primary features will be developed first, whereas secondary features will be developed on available time and resources.

4.1 User Authentication

4.1.1 Description and Priority

The User Authentication component enables a secure log in for users using the DataBridge system. The users should present their credentials to gain access to the patient data or functionality within the system. Once successfully authenticated, the system will determine the user's role and give access to the authorized functions.

This component will be required since the system will hold sensitive information related to the patients. It forms the basis of role-based access control and auditing in the system.

Priority: **High**

4.1.2 Stimulus/Response Sequences

- 1.A.0: The user opens the DataBridge application.
- 1.A.1: The system displays the login page.
- 1.A.2: The user enters their username and password.
- 1.A.3: The user submits the login form.

- 1.A.4: The system verifies the credentials.
- 1.A.5: If the credentials are valid, the system authenticates the user and loads the appropriate interface.
- 1.A.6: If the credentials are invalid, the system denies access and displays an error message.
- 1.A.7: The user may attempt to authenticate again.

4.1.3 Functional Requirements

DB-AUTH-01: The system will require a certain user to authenticate before accessing protected patient information.

DBR-AUTH-02: The system will require a username and password during authentication.

DBR-AUTH-03: The system will verify submitted credentials against the system's user records.

DBR-AUTH-04: The system will deny access when the submitted credentials are invalid.

DBR-AUTH-05: The system will display a clear error message when/if authentication fails.

DBR-AUTH-06: The system will identify the authenticated user's assigned role after successful authentication.

DBR-AUTH-07: The system will prevent unauthenticated users from accessing protected application pages through direct navigation.

4.2 Role-Based Access Control

4.2.1 Description and Priority

Role-Based Access Control (RBAC) is a function that determines which functions and data can be accessed by a user depending on the user's role. Different user roles will be catered for in the project. The roles to be catered for include nurses, charge nurses, and administrators.

RBAC will minimize the chances of users having access to functions or data not within their roles.

Priority: **High**

4.2.2 Stimulus/Response Sequences

- 1.B.0: A user successfully authenticates.
- 1.B.1: The system identifies the user's role.
- 1.B.2: The system loads the appropriate permissions.
- 1.B.3: The user attempts to access a system function.
- 1.B.4: The system checks the user's permissions.
- 1.B.5: If the user has permission, the system allows the requested action.
- 1.B.6: If the user does not have permission, the system denies the action and displays an appropriate message.

4.2.3 Functional Requirements

DBR-RBAC-01: The system will assign each authenticated user a defined role.

DBR-RBAC-02: The system will enforce permissions based on the user's assigned role.

DBR-RBAC-03: The system will provide nurse users access to appropriate bedside documentation functions.

DBR-RBAC-04: The system will provide administrative users access to administrative functions.

DBR-RBAC-05: The system will prevent users from accessing functions that are not permitted for their role.

DBR-RBAC-06: The system will verify authorization before performing protected actions.

DBR-RBAC-07: The system will display/show an authorization error when a user attempts to access a restricted function.

4.3 Bedside Documentation

4.3.1 Description and Priority

Bedside Documentation enables authorized nurses to provide structured data about patients using a web-enabled interface which can be used on a tablet computer. The aim of this feature is to minimize the recording of data outside of this system and then re-entering that data.

The bedside documentation interface as one of the main parts of the system

Priority: **High**

4.3.2 Stimulus/Response Sequences

- 1.C.0: A nurse authenticates into the system.
- 1.C.1: The nurse selects a patient.
- 1.C.2: The system displays the documentation form.
- 1.C.3: The nurse enters the required information.
- 1.C.4: The nurse submits the documentation.
- 1.C.5: The system validates the information.
- 1.C.6: If the information is valid, the system stores the submission.
- 1.C.7: If information is missing or invalid, the system identifies the problem and requests correction.
- 1.C.8: After successful submission, the system provides confirmation.

4.3.3 Functional Requirements

DBR-DOC-01: The system will provide a browser-based interface for bedside documentation.

DBR-DOC-02: The interface will support use on a tablet-sized display.

DBR-DOC-03: The system will allow authorized users to enter structured patient information.

DBR-DOC-04: The system will identify required fields within the documentation form.

DBR-DOC-05: The system will validate required information before accepting a submission.

DBR-DOC-06: The system will identify missing required information to the user.

DBR-DOC-07: The system will prevent invalid submissions from being processed.

DBR-DOC-08: The system will provide confirmation after a successful documentation submission.

DBR-DOC-09: The system will associate each submission with the authenticated user and selected patient.

4.4 Backend Validation and Processing

4.4.1 Description and Priority

The Backend Validation and Processing feature processes data entered through the bedside interface by performing server-side validation before any further processing of the data. It acts as another layer of validation other than browser-based validation.

Validated information will be prepared for the project's interoperability workflow.

Priority: **High**

4.5.2 Stimulus/Response Sequences

- 1.D.0: The user submits documentation.
- 1.D.1: The frontend sends the information to the backend API.
- 1.D.2: The backend receives the request.
- 1.D.3: The backend validates the submitted data.
- 1.D.4: If the data is valid, processing continues.
- 1.D.5: If the data is invalid, the backend rejects the request.
- 1.D.6: The backend returns an appropriate response to the frontend.

4.4.3 Functional Requirements

DBR-BVAL-01: The backend will receive documentation submissions from the frontend.

DBR-BVAL-02: The backend will validate submitted data before processing it.

DBR-BVAL-03: The backend will reject requests containing invalid required data.

DBR-BVAL-04: The backend will return an error response when validation fails.

DBR-BVAL-05: The backend will process successfully validated data.

DBR-BVAL-06: The backend will not process a submission that fails required validation.

4.5 HL7v2 Message Generation

4.5.1 Description and Priority

Message Generation in HL7v2 turns the relevant, validated patient data into an HL7 version 2 message. This message will serve to show how the information moves from the bedside application to an EHR receiving system.

HL7v2 is being used due to the fact that this project targets intercommunication with legacy healthcare systems.

Priority: **High**

4.5.2 Stimulus/Response Sequences

- 1.E.0: The backend receives valid patient information.
- 1.E.1: The system prepares the applicable information for HL7v2 conversion.
- 1.E.2: The system generates an HL7v2 message.
- 1.E.3: The system validates the generated message structure.
- 1.E.4: The system sends the message to the simulated EHR receiver.
- 1.E.5: The receiver accepts and displays the message.

4.5.3 Functional Requirements

DBR-HL7-01: The system will generate an HL7v2 message from applicable validated patient information.

DBR-HL7-02: The system will use the project's defined HL7v2 message structure.

DBR-HL7-03: The system will preserve applicable patient information during message generation.

DBR-HL7-04: The system will send successfully generated HL7v2 messages to the simulated EHR receiver.

DBR-HL7-05: The system will identify an error when an HL7v2 message cannot be generated.

DBR-HL7-06: The system will prevent incomplete or invalid patient data from being converted into a final message.

4.6 Simulated EHR Receiver

4.6.1 Description and Priority

Simulated EHR Receiver is a test environment which receives HL7v2 messages produced by DataBridge. The receiver will show the incoming messages in order to be able to showcase the complete flow of data without needing an active hospital EHR.

Priority: **High**

4.6.2 Stimulus/Response Sequences

- 1.F.0: The backend generates an HL7v2 message.
- 1.F.1: The system sends the message to the simulated EHR receiver.
- 1.F.2: The receiver accepts the message.
- 1.F.3: The receiver processes the message.
- 1.F.4: The receiver displays the received information.
- 1.F.5: If the message cannot be processed, the receiver displays an error.

4.6.3 Functional Requirements

DBR-EHR-01: The simulated EHR receiver will accept HL7v2 messages from the backend.

DBR-EHR-02: The receiver will display successfully received messages.

DBR-EHR-03: The receiver will provide an indication when a message has been received.

DBR-EHR-04: The receiver will identify messages that cannot be processed.

DBR-EHR-05: The system will provide a view for authorized users to inspect received messages.

DBR-EHR-06: The receiver will operate independently from a live hospital EHR.

4.7 Audit Logging

4.7.1 Description and Priority

The Audit Logging capability logs specific actions performed within the system. Every entry in the log will specify the user who performed the action and the time of action. This information will prove useful in verifying activities in the system.

Priority: **Medium**

4.7.2 Stimulus/Response Sequences

- 1.G.0: An authenticated user performs a monitored action.
- 1.G.1: The system records the action.
- 1.G.2: The system records the associated user.
- 1.G.3: The system records the date and time.

- 1.G.4: An authorized user opens the audit log.
- 1.G.5: The system displays the available audit records.

4.7.3 Functional Requirements

DBR-AUDIT-01: The system will record defined security and user activities.

DBR-AUDIT-02: Each audit record will identify the user associated with the activity.

DBR-AUDIT-03: Each audit record will contain the date and time of the activity.

DBR-AUDIT-04: The system provides an audit-log interface to authorized users.

DBR-AUDIT-05: The system prevents unauthorized users from modifying audit records.

DBR-AUDIT-06: The system retains audit records according to the project's defined retention period.

4.8 Document Upload, Classification, and OCR

4.8.1 Description and Priority

This feature enables an authorized user to upload documents and categorize them through document classification and metadata extraction by OCR process. This is done for illustrating purposes of document organization and metadata processing and not for any diagnosis of any medical condition.

These functionalities do not provide any diagnosis of the medical content.

Priority: **Medium**

4.8.2 Stimulus/Response Sequences

- 1.H.0: An authorized user opens the document interface.
- 1.H.1: The user selects a supported document.
- 1.H.2: The system uploads the document.
- 1.H.3: The system processes the document.
- 1.H.4: OCR extracts available metadata when applicable.
- 1.H.5: The system classifies the document.
- 1.H.6: The system displays the classification and extracted metadata.

4.8.3 Functional Requirements

DBR-OCR-01: The system will allow authorized users to upload supported documents.

DBR-OCR-02: The system rejects unsupported file types.

DBR-OCR-03: The system processes supported scanned documents using OCR.

DBR-OCR-04: The system extracts available metadata from documents when supported by the OCR process.

DBR-OCR-05: The system will classify documents for organizational purposes.

DBR-OCR-06: The system will notify the user when document processing fails.

DBR-OCR-07: The system will not use document classification to provide medical diagnoses.

DBR-OCR-08: The system will not provide treatment recommendations based on uploaded documents.

4.9 Secure Document Transfer

4.9.1 Description and Priority

Secure Document Transfer facility enables those who have permission to bundle several documents in one package for transferring to a particular recipient. The facility is meant to provide an example of secure transfer of documents within the DataBridge system.

Priority: **Medium**

4.9.2 Stimulus/Response Sequences

- 1.I.0: An authorized user selects one or more documents.
- 1.I.1: The user chooses the secure transfer option.
- 1.I.2: The system creates a transfer package.
- 1.I.3: The user identifies the intended recipient.
- 1.I.4: The system creates the protected transfer.
- 1.I.5: The recipient attempts to access the package.
- 1.I.6: The system verifies the recipient's authorization.
- 1.I.7: Authorized recipients can access the package.

4.9.3 Functional Requirements

DBR-TRANSFER-01: The system will allow authorized users to select multiple documents for transfer.

DBR-TRANSFER-02: The system will group selected documents into a transfer package.

DBR-TRANSFER-03: The system will associate a transfer package with its intended recipient.

DBR-TRANSFER-04: The system will require recipient authentication before providing access to protected documents.

DBR-TRANSFER-05: The system will deny access when the recipient cannot be authenticated.

DBR-TRANSFER-06: The system will indicate when a secure transfer cannot be completed.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The DataBridge system shall provide responsive performance appropriate for bedside documentation and administrative use. Performance requirements are intended to minimize delays during the nurse's workflow and provide timely feedback when information is submitted or processed.

NFR-PERF-01: The system shall provide a responsive user interface suitable for use during active bedside documentation.

NFR-PERF-02: The system shall provide user feedback after a documentation submission has been received by the backend.

NFR-PERF-03: The backend shall validate submitted documentation before the information is processed or converted into an HL7V2 message.

NFR-PERF-04: The system shall process a valid documentation submission through validation, HL7V2 message generation, and transmission to the simulated EHR receiver without requiring manual intervention.

NFR-PERF-05: The simulated EHR receiver shall display successfully received HL7v2 messages after they have been transmitted by the backend.

NFR-PERF-06: The system's performance shall be evaluated using synthetic patient data during development and testing.

NFR-PERF-07: Specific maximum response-time targets for individual operations shall be established and documented as TBD if required by the final testing plan.

5.2 Safety Requirements

DataBridge is a healthcare documentation and interoperability prototype and is not intended to function as a diagnostic or clinical decision-making system. Safety requirements are intended to prevent the system from presenting itself as a replacement for professional clinical judgment.

NFR-SAFE-01: The system shall not provide medical diagnoses based on patient information or uploaded documents.

NFR-SAFE-02: The system shall not provide treatment recommendations based on patient information or uploaded documents.

NFR-SAFE-03: The document classification and OCR functionality shall be limited to organizational and metadata-extraction purposes.

NFR-SAFE-04: The system shall identify missing or invalid required documentation information before accepting the submission.

NFR-SAFE-05: The system shall prevent documentation that fails required validation from being processed as a completed submission.

NFR-SAFE-06: The system shall provide confirmation after a successful documentation submission so that the user can determine whether the information was accepted.

NFR-SAFE-07: The system shall use synthetic patient data during development, testing, and demonstration of the capstone system.

NFR-SAFE-08: The system shall not require access to a live hospital EHR or live patient information for normal operation of the capstone demonstration.

5.3 Security Requirements

DataBridge shall protect patient-related information and restrict access to authorized users. Security requirements are based on authentication, multi-factor authentication, role-based access control, least privilege, encryption, audit logging, and controlled document sharing.

NFR-SEC-01: The system shall require authentication before a user can access protected application functionality or patient information.

NFR-SEC-02: The system shall require a username and password as part of the user authentication process.

NFR-SEC-03: The system shall use multi-factor authentication as an additional verification step for protected user access.

NFR-SEC-04: The system shall assign an authenticated user a defined role.

NFR-SEC-05: The system shall enforce permissions according to the user's assigned role.

NFR-SEC-06: The system shall follow the principle of least privilege by limiting users to the functions and information required for their assigned role.

NFR-SEC-07: The system shall prevent unauthenticated users from accessing protected application pages or patient information.

NFR-SEC-08: The system shall prevent users from accessing functions or information outside of their assigned permissions.

NFR-SEC-09: The system shall encrypt sensitive information while it is transmitted between system components.

NFR-SEC-10: The system shall protect sensitive information stored by the application through encryption or another approved protection mechanism.

NFR-SEC-11: The system shall record defined security and user activities in an audit log.

NFR-SEC-12: Audit records shall identify the user associated with the activity and the date and time of the activity.

NFR-SEC-13: The system shall restrict access to audit-log information to authorized users.

NFR-SEC-14: The system shall prevent unauthorized users from modifying audit records.

NFR-SEC-15: Secure document-sharing links shall require recipient authentication before protected documents can be accessed.

NFR-SEC-16: Secure document-sharing links shall expire after the defined expiration condition is reached.

NFR-SEC-17: The system shall record access to secure document-sharing links in the audit log.

NFR-SEC-18: The system shall use synthetic patient data for development, testing, and demonstration and shall not require real patient information for the capstone implementation.

NFR-SEC-19: The system shall be designed using HIPAA-aligned security and privacy principles; however, the capstone implementation shall not be represented as a production HIPAA-certified or HIPAA-compliant hospital system.

5.4 Software Quality Attributes

Usability

NFR-QUAL-01: The nurse-facing interface shall provide a simple workflow for selecting a patient, entering bedside information, submitting the information, and receiving confirmation.

NFR-QUAL-02: The nurse-facing interface shall be suitable for use on a tablet-sized display.

NFR-QUAL-03: The system shall provide clear feedback when required information is missing, invalid, or successfully submitted.

Reliability

NFR-QUAL-04: The system shall reject invalid documentation rather than processing it as valid patient information.

NFR-QUAL-05: The system shall provide an indication when an HL7v2 message cannot be generated or transmitted successfully.

NFR-QUAL-06: The simulated EHR receiver shall identify messages that cannot be processed.

Interoperability

NFR-QUAL-07: The system shall use HL7V2 as the interoperability message format for communication between the DataBridge backend and the simulated EHR receiver.

NFR-QUAL-08: The system shall preserve applicable patient information when converting validated documentation into an HL7v2 message.

Maintainability

NFR-QUAL-09: The system shall maintain separation between the browser-based frontend, backend API, interoperability/message-generation component, and simulated EHR receiver.

NFR-QUAL-10: The implementation shall use documented and identifiable third-party libraries and frameworks so that project dependencies can be maintained throughout development.

Portability

NFR-QUAL-11: The system shall operate through a standard web browser without requiring installation of a native mobile application.

NFR-QUAL-12: The system shall support tablet-based bedside use and standard desktop-browser use for administrative and development activities.

Testability

NFR-QUAL-13: The system shall support end-to-end testing using synthetic patient data.

NFR-QUAL-14: The system shall allow the documentation workflow to be tested independently of a live hospital EHR through the simulated EHR receiver.

NFR-QUAL-15: The system shall provide sufficient system feedback and audit information to verify successful and unsuccessful user actions during testing.

5.5 Business Rules

The following business rules define the access and operating principles of the DataBridge system.

BR-01: A user must successfully authenticate before accessing protected DataBridge functionality.

BR-02: Each authenticated user shall have an assigned role.

BR-03: Nurse users shall be permitted to perform authorized bedside documentation functions for patients assigned to them.

BR-04: Charge nurses shall have access to the broader unit-level patient census and applicable unit information and alerts.

BR-05: Administrators shall have access to administrative functions, including user management and authorized audit-log review.

BR-06: Users shall not receive access to functions or information outside the permissions associated with their assigned role.

BR-07: Patient documentation shall be associated with the authenticated user who submitted it and the selected patient.

BR-08: Audit records shall identify the user and time associated with monitored system activities.

BR-09: Document classification and OCR shall be used for organizational and metadata purposes only and shall not be used to provide medical diagnoses or treatment recommendations.

BR-10: Development and demonstration of the capstone system shall use synthetic patient data rather than real patient information.

BR-11: The simulated EHR receiver shall be used for interoperability demonstration and testing in place of a live hospital EHR.

BR-12: Secure document-sharing access shall be limited to the intended and authenticated recipient and shall terminate when the defined expiration condition is reached.

Appendix A: Glossary

API (Application Programming Interface): A mechanism that allows different software components to communicate with one another.

Audit Log: A record of system activities that identifies actions, users, and the time associated with those actions.

Backend: The server-side portion of DataBridge responsible for receiving, validating, processing, and transmitting information.

DataBridge: The browser-based bedside documentation system described by this SRS.

EHR (Electronic Health Record): A digital system used to store and manage patient healthcare information.

FastAPI: A Python framework used to develop the DataBridge backend API.

FHIR (Fast Healthcare Interoperability Resources): A healthcare interoperability standard. It is not the primary messaging standard selected for this project.

HL7v2 (Health Level Seven Version 2): A healthcare messaging standard used by DataBridge to represent and transmit patient-related information to the simulated EHR receiver.

HIPAA (Health Insurance Portability and Accountability Act): A U.S. law containing requirements related to healthcare information privacy and security.

OCR (Optical Character Recognition): Technology used to extract text or metadata from supported scanned documents.

PHI (Protected Health Information): Individually identifiable health information that is protected under applicable healthcare privacy requirements.

RBAC (Role-Based Access Control): An access-control method that determines system permissions according to a user's assigned role.

Simulated EHR Receiver: The test component created by the project team to receive and display HL7v2 messages without connecting to a live hospital EHR.

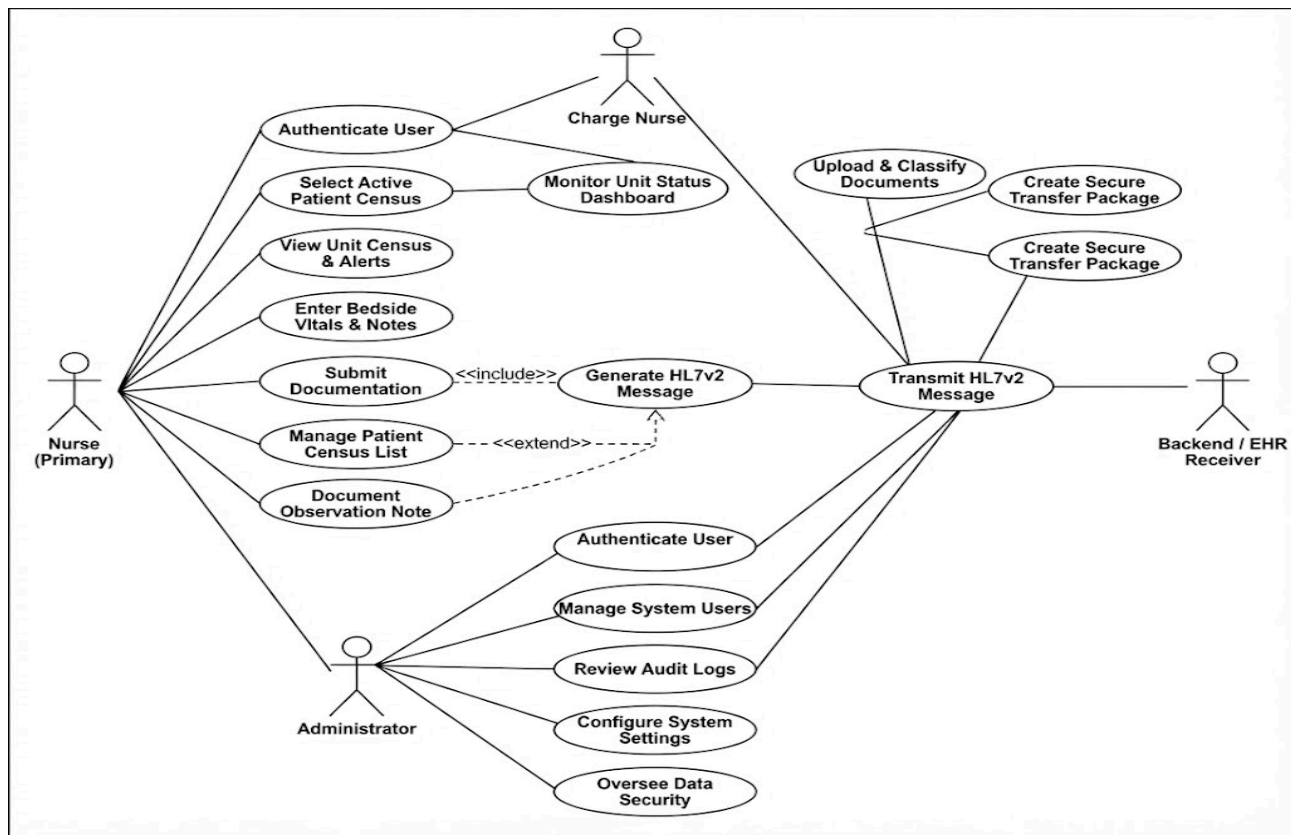
Synthetic Data: Artificial data created for development, testing, or demonstration rather than obtained from real patients.

Frontend: The browser-based portion of DataBridge that users interact with to authenticate, select patients, and enter documentation.

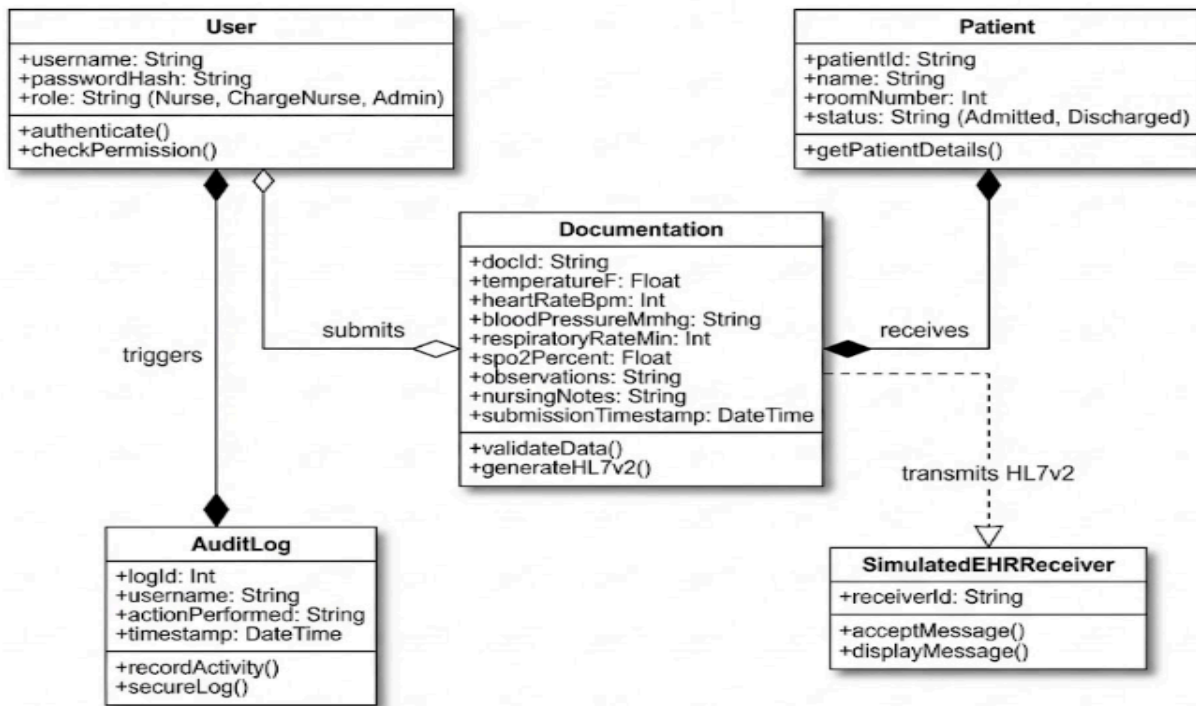
MFA (Multi-Factor Authentication): An authentication method that requires an additional verification step beyond a username and password.

One-Time Sharing Link: A temporary protected link used to provide controlled access to shared documents and that expires after its defined expiration condition.

Appendix B: Use-Case Main Structure



Appendix C: UML Model



Appendix D: To Be Determined List

Determined items during the development and testing of DataBridge.

1. Final database technology and deployment setup.
2. Final response time requirements for individual system operations.
3. Final expiry time for secure document-sharing links.
4. Final maximum usage limit for temporary document-sharing links, if any.
5. Final HL7v2 messages and segments required for demonstration.
6. Final multi-factor authentication implementation process.
7. Final optical character recognition (OCR) tool/library to be used for metadata extraction from documents.
8. Final document file formats supported by document upload function.
9. Final audit log storage period for capstone implementation.
10. Final deployment and hosting setup for the project demonstration.
11. Final test cases and criteria for end-to-end system testing.
12. Final user interface changes resulting from testing and project feedback.